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END TERM EXAMINATION

SECOND SEMESTER [B.TECH] JUNE 2024

Paper Code: ES-102**Subject: Programming in 'C'****Time: 3 Hours****Max. Marks: 60**

Note: Attempt five questions in all including Q. No. 1 which is compulsory. Select one question from each unit.

Q. No.	Question	Marks
Q1	Attempt all questions:	[5×4=20]
(a)	Difference between while and do-while loop?	[4]
(b)	Define different types of sorting?	[4]
(c)	Define pre-processor and its usage in programming?	[4]
(d)	Explain the basic structure and compilation process of C program?	[4]
(e)	Difference between structure and union?	[4]
UNIT - I		
Q2 (a)	What is a token? Explain the different types of tokens like keyword, identifiers, constants, strings with examples?	[5]
(b)	Difference between flowchart and algorithm? Write an algorithm and flowchart to find the factorial of a number?	[5]
Q3 (a)	Define Operator? Explain with example different types of operator: arithmetic, logical, bitwise and ternary operator?	[6]
(b)	Find the output of the following program: <pre>int main() { char str1[20] = "Hello", str2[20] = "World"; printf("%s\n", strcpy(str2, strcat(str1, str2))); printf("%s\n", strrev(str1)); printf("%d\n", strlen(str2)); return 0; }</pre>	[4]
UNIT - II		
Q4 (a)	Define Recursion function? Write a C program to find the Fibonacci series using recursion?	[5]
(b)	Define an array? Write the advantages of an array? Write a C program to transpose of a matrix?	[5]
Q5 (a)	Explain call by value and call by reference with example?	[5]
(b)	Define string? How it can be declared and initialized? Explain any five string function with example?	[5]
UNIT - III		
Q6 (a)	Explain File Handling? What is the need for file handling? Discuss file access modes for text files: r, a, w, r+, a+, w+ ?	[5]
(b)	What is the need of a structure? How will you access the members of a structure? Write a program to read and print the details of a person using Structure?	[5]
Q7 (a)	Define Pointer with an example? How to access a variable through pointer? Write a C program to add two variables using Pointer?	[5]
(b)	Define File Handling and explain its operations?	[5]
UNIT - IV		
Q8 (a)	What is Binary search? Search item 15 from the following sorted data elements using binary search: 2, 5, 8, 10, 15, 24, 38, 59, 75, 90	[5]

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Q. No.	Question	Marks
(b)	What is insertion sort? Write a C program to implement insertion sort?	[5]
Q9 (a)	What is header file? Explain library functions of the following header files in detail: time.h , ctype.h , setjmp.h , string.h , stdarg.h , unistd.h , stdio.h , assert.h , stdlib.h ?	[5]
(b)	Define Selection Sort? Sort the following elements using Selection Sort: 77, 33, 44, 11, 88, 22, 66, 55	[5]

**** End of Question Paper ****